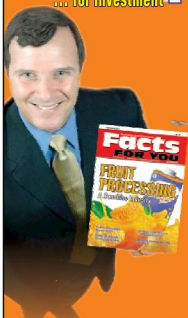


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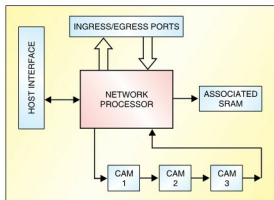


Fig. 3: A typical NPU board depicting the various units that support the functionality

with features to directly link them to the SRAM in order to avoid the path back to the network processor and hence save some time.

Applications of CAMs

1. **Quality of service (QoS) through a CAM.** CAMs generally have an associated SRAM. This SRAM contains routing information and QoS related data. This data helps the network processor to decide the fate of the received packet. Some of the QoS services include:

- (a) Packet monitoring
- (b) Billing
- (c) Firewall
- (d) Content protection

2. **Implementation of ARP.** When a new host gets associated with a network, the foremost thing that the network processor does is to update its network database with the MAC and IP addresses of the new hosts. This information update can happen using one of the very popular protocols of TCP/IP called the address resolution protocol and the new addresses get updated on the CAM table.

3. **IP forwarding address translation (through ACLs).** When a packet arrives at the Ingress port of the network processor, it has to decide on which port (among the many available) the data packet should be forwarded. ACLs also contain information regarding the rules that apply on a particular IP address, such as permission for communication between two specific

IP addresses, and encryption of data during communication between two hosts or a host and a server. This information is stored in the associated SRAM.

4. **MPLS switching at core and edge routers.** MPLS switching involves label look-up in place of an IP address look-up. MPLS labels consist of label value, QoS, stack-flag and TTL (time to live). When a look-up for such a

data happens, the network processor can use the relevant information and decide the fate of the packet; i.e., if the TTL has lapsed, it would drop the packet and so on.

5. **Signature recognition.** With the performances and speeds (up to 10 Gbps) that current technologies are achieving in communication, the performance of the CAM is extremely important. One of the foremost applications that the Internet needs is data protection. With new ways coming in vogue to hack an endpoint, the need for protection from such attacks, be it through a virus or a spam, is necessary. If such a process can be achieved in hardware, it improves the throughput drastically. CAMs can help in such a cause by maintaining a signature database and an associate processor (if it can be built to work at line speeds) can scan the packets for such signatures and drop packets if necessary; such a functionality would add immense value to the content moved across the Internet.

Way forward

CAMs, as seen above, are becoming an important tool in the communications arena and any research towards further improving their value by enhancing their throughput or reducing the cost would add to the advantage. ●

The author is working as a design engineer at NeltLogic Semiconductor